

REVISED TESTING PLAN

Web-Based Personnel Firearms Records, Notification Monitoring and Renewal System with Analytics

4th Infantry Division - 10th Field Property Accountability Office (APAO)

Document version	2.0 - system-aligned revision
Baseline reviewed	Laravel application in apao_renewal workspace
Source plan	TESTING PLAN.pdf (16 pages)
Review date	01 October 2026
Status	Ready for controlled execution; results not pre-filled

Purpose. This revision preserves the four-level structure of the supplied plan, corrects inaccurate expectations, and adds coverage for implemented security, recovery, export, notification, session, and document-generation behavior.

Important. This is a test plan, not a test report. Blank Actual Result, Status, Evidence, Tester and Date fields must be completed during execution. A planned test may expose a defect; it must not be marked Pass merely because it appears in this document.

1. Revision assessment

The supplied plan had a useful functional backbone, but it did not fully match the current application. The following corrections are incorporated throughout this revision.

Finding	Correction in this revision	Impact
UT-101 treated staff@apao.mil.ph and admin@apao.mil.ph as invalid.	Use genuinely malformed inputs (for example staff@ or staff.apao.mil.ph); valid domain-style addresses must pass syntax validation.	Prevents a false expected result.
Password policy was described as one rule.	Separate 10-character admin/profile rules from the 12-character password-reset and super-admin rules.	Matches implemented validators.
RPCSP export was called CSV.	Rename to Excel-compatible .xls and verify workbook structure and binary-data exclusion.	Matches the UI and JavaScript exporter.
ISO/IEC 25010 section omitted Compatibility and called every subtotal a SUS score.	Use nine ISO/IEC 25010:2023 characteristics and keep the 10-item SUS instrument separate.	Corrects evaluation method and terminology.
Security and recovery flows were incomplete.	Add CAPTCHA, OTP reset, security headers, injection handling, cache controls, role boundaries, session timeout and single-tab cases.	Covers implemented protections.
Core workflow details were absent.	Add uniqueness pre-checks, renewal-date boundaries, notification read state, inspection/PAR PDFs, PAR metrics/history, and audit/report exports.	Improves traceability to current routes and client logic.
Automated expiry notification was assumed complete.	Classify scheduled expiry notification as a known implementation gap until the command and scheduler are implemented.	Avoids accepting an unavailable capability.

Implementation gaps to track as defects or deferred requirements: (1) `app:notify-expired-personnel` has an empty handler and no recurring schedule; (2) the public `/send-mail` route requires an authorization/security decision; (3) admin personnel create/update endpoints do not apply the same server-side validation as staff registration; (4) several legacy controller methods use camelCase columns and are not the active routes. These are test targets, not assumed passes.

2. Test governance and controls

2.1 Scope

In scope: authentication, account recovery, role/session controls, user management, personnel and firearm records, archive, ICS inspection, renewal history, PAR lifecycle, notifications/email, dashboards, reports/RPCSP, audit logging, PDF/Excel outputs, and representative non-functional quality checks.

Out of scope unless separately authorized: production penetration testing, destructive production data tests, mail-provider load testing, disaster recovery of infrastructure, and formal certification against ISO standards.

2.2 Entry and exit criteria

Gate	Required condition
Entry	Approved build identified by commit/tag; test database backup available; migrations applied; Admin, Staff and Super Admin test accounts available; mail is sandboxed; supported browsers/devices declared.
Execution	Each test records build, environment, preconditions, actual result, status, evidence reference, tester and date. Failed cases receive a defect ID and severity.
Exit	100% of Critical/High cases executed; no open Critical defect; High defects accepted by owner or fixed/retested; core workflow pass rate at least 95%; UAT acceptance signed with exclusions listed.

2.3 Test environments and data

Item	Minimum setup
Application	Dedicated test instance using the candidate build, production-like configuration and HTTPS where HSTS is evaluated.
Database	Isolated seeded database with normal, boundary, duplicate, archived, expired, within-renewal, inspected and PAR-linked records.
Accounts	1 Super Admin, 2 Admin (to test last-admin safeguards), 2 Staff, 1 inactive account; unique emails and known strong passwords.
Mail	Test mailbox or mail trap; verify recipient, subject, content, delivery attempt, error handling and audit event without contacting real personnel.
Browsers	Current organization-supported desktop browsers plus one supported mobile viewport. Record exact versions.
Evidence	Screenshots, downloaded file, database query/output, audit-log row, mail-trap record and automated-test output as applicable.

2.4 Severity and status

Severity	Definition	Examples
Critical	Security breach, unrecoverable data loss, or core workflow unavailable.	Unauthorized firearm-record access; renewal corrupts records.
High	Major function fails with no acceptable workaround.	Cannot inspect, renew, issue PAR, or authenticate valid users.
Medium	Function is incorrect but workaround exists.	Wrong filter/export value; notification read state fails.
Low	Cosmetic, wording or minor usability issue.	Misaligned label; non-blocking responsive defect.

3. Standard execution record

Use this block for every automated or manual test. Do not replace the expected result with the observed result.

Field	Value to record	Field	Value to record
Test ID		Build/commit	
Environment		Browser/device	
Tester		Execution date/time	
Actual result		Status	Not Run / Pass / Fail / Blocked
Evidence reference		Defect ID / severity	

Test data rules

Data class	Examples and boundaries
Identity	Mixed-case email normalization; duplicate email/contact/AFP serial/pistol serial; Unicode/invisible characters; maximum lengths.
Dates	Today, yesterday, 30 days before validity, validity date, leap-day birth, missing birth date, inverted report range.
Inspection	All serviceable/N/A; Repair/Damaged/Missing; Replace; Unserviceable; mixed findings to verify priority.
Files/data URLs	Valid photo/signature; omitted optional image; oversized/malformed payload; verify binary content is excluded from exports.
Adversarial	SQL metacharacters as data; invalid arrays instead of scalars; expired/reused OTP; wrong role; stale session; repeated login/notify requests.

4. Level 1 - Unit and component testing

Automate pure rules and request validation where practical. Existing automated tests are evidence candidates, but execution status remains blank until rerun on the candidate build.

ID	Target	Scenario / data	Expected result
UT-101	Email syntax and normalization	Accept valid staff@apao.mil.ph; reject staff@ and staff.apao.mil.ph; trim/lowercase where implemented.	Valid syntax passes; malformed syntax fails with field error.
UT-102	10-character password policy	Exercise Admin-created/profile passwords at 9 and 10+ chars with mixed case, number and symbol.	Under 10 or missing class fails; compliant value passes.
UT-103	12-character reset/owner policy	Exercise OTP reset and super-admin creation at 11 and 12+ chars.	Under 12 or missing class fails; compliant value passes.
UT-104	Password hashing/check	Store a valid password and compare correct/incorrect plaintext.	Only hash is persisted; check succeeds only for correct value.
UT-105	CAPTCHA challenge	Generate arithmetic challenge; submit correct, incorrect and reused answers.	Correct single-use answer proceeds; incorrect/reused answer returns 422 and a fresh challenge.
UT-106	Login throttling	Submit five credential failures for same normalized email/IP, then another attempt.	Lock activates for about five minutes; 429 and retry information returned.
UT-107	Inactive and unknown roles	Authenticate inactive user and account with unsupported role.	Access denied without dashboard session.
UT-108	Role authorization	Staff requests Admin resource; Super Admin requests Admin and Staff resources.	Staff denied/redirected; Super Admin may use Admin area but not Staff area.
UT-109	Session invalidation/cache	Logout then query session status and request protected content.	Session/token invalidated; 401/redirect; no-store headers present.
UT-110	Profile change safeguards	Change email with wrong/missing current password; reuse another email; change password to same value.	Request rejected and stored/session data unchanged.
UT-111	Personnel uniqueness	Check and submit duplicate email, contact, AFP serial and pistol serial with case variations.	Availability and final write reject duplicates; race returns controlled 422.
UT-112	Renewal status boundary	Evaluate dates past validity, on validity, within 30 days, and over 30 days away.	Expired only after validity; within-renewal starts 30 days before; otherwise valid.
UT-113	Renewal validity date	Renew on ordinary birthday, leap-day birthday in non-leap target year, and missing birthday.	Birthday two calendar years after renewal; Feb 29 clamps to Feb 28; missing DOB adds two years.
UT-114	RPCSP condition priority	Mix Serviceable/N/A, repair group, Replace and Unserviceable findings.	Unserviceable > For Replacement > For Repair > Serviceable.
UT-115	Report date filter	Inclusive date bounds, timestamp normalization, no date, invalid date and From > To.	Correct rows returned; invalid/inverted ranges rejected.
UT-116	Single-tab ownership	Parse live, expired, malformed and same-tab ownership records.	Only a different live tab is a conflict; tab ID stays stable during same-tab navigation.
UT-117	Security headers	Create HTTP and HTTPS responses.	nosniff, SAMEORIGIN, referrer and permissions policies always; HSTS only over HTTPS.
UT-118	PAR input rules	Boundary quantities/costs/dates, equipment count, reason length and signatures.	Invalid bounds fail; valid update/replacement payload passes.

5. Level 2 - Integration testing

ID	Interfaces	Scenario	Expected cross-module behavior
IT-101	Login UI <-> session/roles	Valid Admin, Staff and Super Admin complete CAPTCHA login.	Session regenerates, last login/audit updates, correct dashboard opens.
IT-102	Forgot password <-> OTP/mail/users	Request OTP, verify before/after 10 minutes, reset with valid/invalid code.	Generic request response; valid code permits strong reset; used/expired code fails; OTP is deleted after reset.
IT-103	Manage Users <-> security/audit	Create/update role/status/password with and without current Admin password.	Sensitive action requires confirmation, hash persists, safeguards and audit entries apply.
IT-104	Registration <-> availability/DB	Run pre-check then register; repeat identifiers and simulate competing submission.	Unique record commits atomically; duplicates return field/control error; item number remains unique.
IT-105	Registration <-> notifications/audit	Staff registers valid personnel with photo/signature.	Personnel row, staff audit event and unread Admin notification are consistent.
IT-106	ICS <-> Admin Inspection	Staff sends eligible record for inspection.	ics_status becomes under; Admin list and notification show the same target.
IT-107	Inspection UI <-> DB	Save each checklist category with remarks and signatories.	New inspection row preserves all selected values and metadata.
IT-108	Inspection <-> RPCSP	Save mixed conditions and reload Admin/Staff reports.	Latest inspection drives one priority-correct RPCSP remark everywhere.
IT-109	Re-inspection <-> renewal guard	Correct defective item and save a new all-serviceable inspection.	History is preserved; latest condition becomes Serviceable; renewal action becomes available.
IT-110	Renewal <-> history/transaction	Mark serviceable firearm for renewal.	Personnel dates/status, inspection next date, renewal history and renewal transaction commit consistently.
IT-111	Renewal <-> notification	Admin notifies Staff after renewal readiness.	Correct role notification is unread, target-linked and auditable.
IT-112	Personnel <-> PAR issuance	Issue first PAR for eligible renewed personnel.	Active PAR, personnel link, signature reuse, status update and audit event agree.
IT-113	PAR replacement chain	Replace an active PAR, then replace the replacement.	Exactly one newest active PAR; all predecessors retained with reason/reference.
IT-114	PAR/inspection <-> PDF	Download PAR and inspection PDFs.	Correct source data appears; document opens, prints and contains no clipped/blank critical fields.
IT-115	Reports <-> latest records	Filter renewal report and open RPCSP after updates.	Current personnel, PAR, renewal and latest-inspection values are used.
IT-116	Reports <-> Excel	Export personnel, renewal, RPCSP and audit data.	Readable .xls downloads match displayed filters; no photo/signature binary data.
IT-117	Notification read state	Open Admin/Staff notifications and mark read.	Unread count and appropriate role flag update without corrupting other role state.
IT-118	Actions <-> audit	Perform login failures, profile/user/personnel/inspection/PAR/email actions.	Audit row captures actor, role, action, target, details, IP and time without secrets.
IT-119	Audit filters <-> query builder	Submit ordinary filters, SQL-like text, arrays and invalid dates.	Bound text is treated as data; non-scalar/invalid dates return 422; database unchanged.
IT-120	Session UI <-> logout endpoint	Allow inactivity countdown, stay signed in, open second tab, use Back after logout.	Role timeout is enforced, active tab owns session, second tab fails closed, stale pages cannot reopen.

6.1 Series 100 - Authentication, recovery, accounts and security

ID	Module	Execution procedure / data	Expected system response
ST-101	Authentication	Open Login; solve CAPTCHA; use valid Admin/Staff/Super Admin credentials.	Role dashboard opens, session changes ID, login audit and last-login time update.
ST-102	Authentication	Use existing email with wrong password and valid CAPTCHA.	Generic invalid message, remaining attempts and fresh CAPTCHA; no account disclosure.
ST-103	Authentication	Enter wrong/reused/missing CAPTCHA.	422; credentials are not accepted; new challenge returned.
ST-104	Authentication	Fail credentials five times, then retry before/after lock period.	429 during lock; login becomes available after expiry.
ST-105	Authentication	Deactivate user and attempt login; alter stored role to unsupported value.	Access refused and no privileged dashboard is exposed.
ST-106	Authorization	Staff manually opens Admin URLs/APIs; Super Admin opens both role areas.	Staff denied; Super Admin enters Admin only and is redirected away from Staff.
ST-107	Session	Remain idle: Admin/Super Admin 15 min, Staff 30 min; test 60-second warning and Stay Logged In.	Warning timing and forced logout match role; activity extends deadline.
ST-108	Session	Open a second authenticated tab for same account.	Conflicting tab logs out/fails closed; single active tab remains.
ST-109	Session	Logout then use Back/Forward and call /session/status.	Protected content does not reappear; status is 401; no-store headers prevent stale use.
ST-110	Password recovery	Request OTP for known and unknown email; repeat 4 times/minute.	Same generic response for account existence; fourth request throttled; mail only for known account.
ST-111	Password recovery	Use wrong, expired and valid six-digit OTP; reset with weak/strong password.	Only valid unexpired OTP plus 12-character policy resets; OTP removed after success.
ST-112	Profile	Update name; change email with/without password; change password.	Session refreshes; uniqueness/current-password/strength rules apply; audit contains no plaintext secrets.
ST-113	Manage Users	Create Admin/Staff with valid/invalid data and correct/incorrect Admin confirmation.	Only confirmed valid account is created with hash and audit record.
ST-114	Manage Users	Deactivate/demote current Admin or last active Admin; reset another user password.	Unsafe change blocked; permitted confirmed change succeeds and is audited.
ST-115	Security headers	Inspect login, dashboard, JSON, HTTP and HTTPS responses.	Configured headers exist; authenticated responses are non-cacheable; HSTS only on HTTPS.
ST-116	Security input	Put SQL-like payloads in audit filters and invisible characters in names/account fields.	Inputs are rejected or treated as data; no expanded result set or database change.

6.2 Series 200 - Personnel and firearm records

ID	Module	Execution procedure / data	Expected system response
ST-201	Personnel registration	Register complete personnel/firearm record with valid image/signature and known issuer.	Unique item number and normalized identifiers save; Admin is notified; record appears across workflows.
ST-202	Personnel registration	Leave required fields blank; future DOB; nonnumeric contact; unknown issuer; negative ammo.	422 with field-specific errors; nothing persists.
ST-203	Personnel registration	Use duplicate email/contact/AFP serial/pistol serial in pre-check and final submission.	Both layers reject duplicate, including case variations; friendly error shown.
ST-204	Personnel registration	Submit two valid registrations concurrently.	Transaction/locking produces distinct item numbers and complete rows.
ST-205	Personnel update	Edit allowed fields, preserve omitted photo/signature, then explicitly replace them.	Values persist and media is preserved/replaced as intended; audit/notification created.
ST-206	Admin personnel validation	Use invalid/duplicate values on Admin create/update.	Expected secure behavior: reject invalid data. Any acceptance is a High defect because these endpoints currently lack parity validation.
ST-207	Archive	Archive active record, restore it, then permanently delete an archived test record with confirmation.	Active/archive lists and archived_at change correctly; permanent removal only after explicit confirmation; actions audited.
ST-208	Search	Search by name, AFP serial and a nonexistent term.	Only matching rows appear; empty state is clear; archived records remain excluded from active views.

6.3 Series 300 - ICS, inspection and renewal

ID	Module	Execution procedure / data	Expected system response
ST-301	ICS	Staff sends eligible personnel for inspection twice.	First send changes status and notifies Admin; repeat is idempotent or produces a controlled response without duplicate corruption.
ST-302	Inspection	Open pending/under record and start checklist.	Correct personnel/firearm fields and checklist display; status reflects in-progress workflow.
ST-303	Inspection	Mark every applicable part Serviceable/N/A and save for renewal.	Overall Serviceable; renewal is permitted.
ST-304	Inspection	Save Repair/Damaged/Missing finding.	RPCSP For Repair; workflow remains under inspection; renewal blocked.
ST-305	Inspection	Save Replace finding, alone and mixed with Repair.	RPCSP For Replacement; renewal blocked.
ST-306	Inspection	Save Unserviceable finding mixed with other defects.	RPCSP Unserviceable (highest priority); renewal blocked.
ST-307	Inspection	Omit required condition/signatory field or submit unsupported condition.	Validation rejects save; no partial inspection/renewal data.
ST-308	Re-inspection	After repair, save a new all-serviceable inspection.	New latest inspection drives reports; prior row remains history.
ST-309	Renewal	Mark latest serviceable firearm for renewal.	Status renewed; date approved/last renewed and birthday-based validity update; history/transaction rows created.
ST-310	Renewal	Renew leap-day DOB and missing DOB records.	Feb 28 target in non-leap year; missing DOB falls back to two years from renewal.
ST-311	Renewal history	Open history after multiple renewals.	Newest-first entries show old/new validity, inspector, remarks and date without overwriting history.
ST-312	Inspection PDF	Download report with and without an inspection row.	A4 PDF opens; current personnel/latest inspection are legible; missing inspection handled gracefully.

6.4 Series 400 - PAR management

ID	Module	Execution procedure / data	Expected system response
ST-401	PAR eligibility	Open issuance list before and after renewal readiness/active PAR.	Only eligible renewed personnel without an active PAR appear; filters/sort/pagination are correct.
ST-402	PAR issuance	Issue first PAR with costs, quantities, dates, equipment and signatories.	Unique PAR becomes Active; personnel becomes par_issued; totals/links/audit are correct.
ST-403	PAR signature reuse	Open document for personnel with stored signature and selected signatories.	Receiver and available signatory signatures appear in correct locations.
ST-404	PAR update	Edit allowed fields and boundary values; attempt valid_until before issued_date.	Valid edit keeps same PAR number; invalid date/bounds return errors.
ST-405	PAR replacement	Replace active PAR with reason <5 chars, then valid reason; replace again.	Short reason fails; valid replacement closes predecessor, creates new active PAR and preserves full chain.
ST-406	PAR PDF	Download/print active and historical PAR documents.	Correct PAR version, personnel/property values, dates, amounts and signatures render without clipping.
ST-407	PAR dashboard	Issue/update/reprint/replace records and inspect dashboard metrics/activity.	Waiting/existing/month/year/replaced metrics and recent actions reconcile with data/audit.

6.5 Series 500 - Dashboards, reports and exports

ID	Module	Execution procedure / data	Expected system response
ST-501	Dashboards	Compare Admin/Staff metric cards with seeded statuses and archived records.	Counts match active records and defined statuses; archived rows excluded.
ST-502	Renewal report	Use inclusive date/status filters, no dates, invalid date and inverted range.	Rows match criteria; invalid/inverted range is blocked with clear feedback.
ST-503	Inspection filters	Select Pending, Under Inspection and Ready for Renewal cards.	Only records matching current workflow/latest inspection state display.
ST-504	RPCSP	Verify Not Yet Inspected, Serviceable, Repair, Replacement and Unserviceable records.	Every row uses latest inspection and priority rule consistently.
ST-505	Excel exports	Export personnel, renewal, RPCSP and audit views after filters.	.xls files open; headings/period/rows match view; special characters escape; binary media excluded.
ST-506	Audit filters/export	Filter by action, user and inclusive dates; exceed 200 matching rows.	Displayed/exported data follows filters; UI documents the 200-row API limit or provides controlled pagination.

6.6 Series 600 - Notifications, email and audit

ID	Module	Execution procedure / data	Expected system response
ST-601	Notifications	Trigger personnel add/update/archive, ICS send, inspection/renewal and email events.	Correct role receives target-linked notification; unread count increments once.
ST-602	Notification read	Mark Admin notifications read and inspect Staff state, then reverse.	Role-specific read flags/counts change correctly; other role state is preserved.
ST-603	Email notify	Send one and bulk selected within/expired reminders; include missing email and mail failure.	Sent/failed/skipped totals are accurate; success audit/notification only after send; controlled error on failure.
ST-604	Scheduled reminders	Run scheduler/command for within-renewal and expired records twice.	Required target: due notices are sent once per rule and audited. Current expected status: Blocked/Fail until command and schedule are implemented.
ST-605	Public mail endpoint	Unauthenticated user requests /send-mail.	Security target: no mail is sent without authorization. Any successful send is a High security defect pending route decision.
ST-606	Audit secrecy	Perform password, OTP, profile and mail actions; inspect log/database/API.	No plaintext password, OTP, session token or full sensitive payload appears in logs/API.

6.7 Series 700 - Non-functional verification

ID	Module	Execution procedure / data	Expected system response
ST-701	Performance	Measure dashboard/list/filter/save/PDF/export with agreed representative volume.	Targets agreed before execution; suggested p95 page/API <=2 s, save <=3 s, PDF/export <=10 s under normal load.
ST-702	Capacity	Load at least agreed peak concurrent users and large record set.	No data corruption; error rate <1%; response targets remain acceptable; resource use recorded.
ST-703	Compatibility	Execute core workflow on declared browsers and supported desktop/mobile widths.	No blocking functional/layout differences; downloads and camera/signature alternatives behave as specified.
ST-704	Accessibility	Keyboard-only navigation, visible focus, labels, error association, contrast and zoom to 200%.	Core tasks remain operable and understandable; critical WCAG failures are logged.
ST-705	Reliability	Interrupt request/network during save; retry; test duplicate submission.	Atomic operations avoid partial/duplicate records; user gets recoverable feedback.
ST-706	Backup/recovery	Restore a sanitized database backup and validate sample personnel, inspection, renewal, PAR and audit links.	Restore meets agreed RPO/RTO and referential integrity checks.

7. Requirements traceability summary

Capability / risk	Unit	Integration	System
Authentication and authorization	UT-101 to UT-110	IT-101 to IT-103, IT-120	ST-101 to ST-116
Personnel/firearm record management	UT-111	IT-104, IT-105	ST-201 to ST-208
Inspection and renewal	UT-112 to UT-114	IT-106 to IT-111	ST-301 to ST-312
PAR lifecycle	UT-118	IT-112 to IT-114	ST-401 to ST-407
Reports, analytics and exports	UT-115	IT-115, IT-116	ST-501 to ST-506
Notifications, email and audit	UT-117	IT-117 to IT-119	ST-601 to ST-606
Session/browser controls	UT-116, UT-117	IT-120	ST-107 to ST-109, ST-703
Performance/reliability/recovery	-	transaction cases	ST-701, ST-702, ST-705, ST-706

The execution owner should add requirement IDs from the approved Software Requirements Specification when available. A test without a requirement may still be valid as a risk-based control; a requirement without a test must be justified.

8. Level 4 - User acceptance testing

UAT validates fitness for APAO work; it does not replace technical security, performance or recovery testing. Participants should complete representative tasks before rating quality.

8.1 Participant profile

Participant ID	Role / position	Organization / unit	Experience	Date	Signature	
UAT-EX-01	Domain expert / client					
UAT-EX-02	Domain expert / client					
UAT-IT-01	IT / security reviewer					
UAT-EU-01	Admin end-user					
UAT-EU-02	Staff end-user					
UAT-EU-03	Staff end-user					

Add rows as needed. Report the actual sample; do not claim a fixed number as an ISO requirement.

8.2 Task-based acceptance scenarios

ID	Role	Task	Acceptance observation
UAT-T01	Staff	Register personnel, verify duplicate warning, send ICS for inspection.	Record is accurate; Admin receives request.
UAT-T02	Admin	Inspect firearm with a defect, save, then review RPCSP.	Condition and report wording match domain expectation.
UAT-T03	Admin	Re-inspect corrected firearm and mark it for renewal.	History is preserved and new validity is correct.
UAT-T04	Staff	Issue PAR, view signature, download document, then replace it.	Documents are complete and replacement chain is understandable.
UAT-T05	Admin/Staff	Filter renewal records and export the result to Excel.	Displayed and exported values match.
UAT-T06	Staff	Send a renewal reminder and review notifications.	Recipient/message/read state meet office workflow.
UAT-T07	Admin	Create/update a Staff account and review the audit trail.	Safeguards are clear and evidence is traceable.
UAT-T08	All	Use profile, theme/navigation and allow inactivity warning to appear.	Interface remains readable and session warning is understandable.

8.3 ISO/IEC 25010:2023-aligned product-quality survey

Scale: 5 Strongly Agree, 4 Agree, 3 Neither Agree nor Disagree, 2 Disagree, 1 Strongly Disagree, N/A Not Observed. ISO alignment structures the questions; it does not by itself certify the product.

The 2023 product-quality model has nine characteristics. This revision therefore adds Compatibility and Safety, renames usability-oriented product questions as Interaction capability, and replaces Portability with Flexibility. The characteristic score is the arithmetic mean of answered items; N/A is excluded.

Functional suitability

Functional suitability - statement	5	4	3	2	1	N/A	Comment / evidence
The system provides the functions needed for personnel, firearm, inspection, renewal and PAR work.							
Saved and displayed information remains correct across related modules.							
Reports, notifications and generated documents support the intended office tasks.							

Performance efficiency

Performance efficiency - statement	5	4	3	2	1	N/A	Comment / evidence
Common pages, searches and filters respond within an acceptable time.							
Saving records and generating PDF/Excel outputs completes without unreasonable delay.							
The system remains responsive at the expected record and user volume.							

Compatibility

Compatibility - statement	5	4	3	2	1	N/A	Comment / evidence
The system works in the browsers and devices approved by the office.							
PDF, Excel and email outputs can be used by the intended supporting tools.							
The application co-exists with normal workstation tools without disrupting them.							

Interaction capability

Interaction capability - statement	5	4	3	2	1	N/A	Comment / evidence
Labels, navigation, feedback and error messages make the next action clear.							
Users can complete assigned tasks with limited assistance and recover from input errors.							
Keyboard use, focus, readability and responsive layout support the intended users.							

Reliability

Reliability - statement	5	4	3	2	1	N/A	Comment / evidence
The system consistently completes normal workflows without unexpected interruption.							
Retries, duplicate submissions and temporary connection problems do not corrupt records.							
Stored personnel, inspection, renewal and PAR relationships remain dependable.							

Security

Security - statement	5	4	3	2	1	N/A	Comment / evidence
Authentication, roles and session controls prevent unauthorized access.							
Sensitive credentials and personal/firearm information are not unnecessarily exposed.							
Important actions are traceable without logging passwords, OTPs or session secrets.							

Maintainability

Maintainability - statement	5	4	3	2	1	N/A	Comment / evidence
The system can be tested and faults can be isolated by qualified technical personnel.							
Changes to one module can be made with limited unintended effects on other modules.							
Configuration, logs, migrations and automated tests support controlled maintenance.							

Flexibility

Flexibility - statement	5	4	3	2	1	N/A	Comment / evidence
The system can adapt to supported device sizes, deployment settings and changing record volume.							
Reports, rules and workflow details can be modified when office requirements change.							
The solution can be installed or migrated to an approved environment with controlled effort.							

Safety

Safety - statement	5	4	3	2	1	N/A	Comment / evidence
Warnings and confirmations reduce the risk of accidental archive, deletion, replacement or renewal.							
The system prevents unsafe renewal when inspection findings are not serviceable.							
Failures produce clear, controlled outcomes instead of silently creating incorrect official records.							

8.4 System Usability Scale (SUS) - separate instrument

Use the same 1-5 response scale. SUS is scored only from these ten alternating positive/negative items: odd-item contribution = response - 1; even-item contribution = 5 - response; sum contributions and multiply by 2.5. Do not label ISO characteristic means as SUS scores.

No.	Statement	1	2	3	4	5	Comment
1	I think that I would like to use this system frequently.						
2	I found the system unnecessarily complex.						
3	I thought the system was easy to use.						
4	I think that I would need the support of a technical person to use this system.						
5	I found the functions in this system were well integrated.						
6	I thought there was too much inconsistency in this system.						
7	I would imagine that most intended users would learn to use this system very quickly.						
8	I found the system very cumbersome to use.						
9	I felt very confident using the system.						
10	I needed to learn a lot of things before I could get going with this system.						

Participant ID	Raw contribution sum (0-40)	SUS score (0-100)	Notes

9. Defect, risk and decision logs

9.1 Defect log

Defect ID	Test ID	Summary	Severity	Owner	Status	Retest evidence

9.2 Known gap / requirement decision log

ID	Topic	Current observation	Required decision / acceptance
GAP-01	Scheduled renewal reminders	Command handler is empty and no recurring schedule is registered.	Implement and retest, or formally remove 'automated' claim from scope/title.
GAP-02	Public send-mail route	Route is outside authenticated groups.	Remove/protect it or document a justified non-sensitive purpose; verify no unauthorized mail.
GAP-03	Admin personnel validation	Admin create/update paths do not mirror Staff server validation.	Add parity validation/uniqueness or accept documented risk; execute ST-206.
GAP-04	Performance targets	No approved SLA/volume baseline found.	Product owner approves record volume, concurrency and response thresholds before ST-701/ST-702.
GAP-05	Backup/restore	Application repository does not define operational backup/RPO/RTO evidence.	Infrastructure owner supplies procedure and targets before ST-706.

10. Test summary and formal acceptance

Metric	Planned	Passed	Failed	Blocked	Not run
Level 1 - Unit/component	18				
Level 2 - Integration	20				
Level 3 - System	61				
Level 4 - UAT tasks	8				

Release recommendation: ☐ Accept ☐ Accept with listed conditions ☐ Reject / retest required

Conditions, exclusions and open risks:

Role	Printed name	Signature	Date
QA / Test Lead			
System Owner / Client			
IT / Security Reviewer			
Project Adviser			

Acceptance statement. By signing, the approver confirms that the recorded test evidence and open risks have been reviewed. Signature does not certify unexecuted tests or waive undisclosed defects.

References

Application baseline: routes, controllers, middleware, models, migrations, views, JavaScript utilities and automated tests in the reviewed workspace.

ISO/IEC 25010:2023, Systems and software engineering - Systems and software Quality Requirements and Evaluation (SQuaRE) - Product quality model. Official overview:
<https://www.iso.org/standard/78176.html>

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